Alejandro Saenz

Engineering Manager, Application Security

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SUMMARY

I lead application security at Twilio and have spent over a decade building and breaking software, across software engineering and product security. I build the automation and metrics that let security teams scale, including Snyk Autofactory and the Multivac security metrics framework. Former lead developer of the open source vulnerability management tool Bulwark; speaker on securing software at scale.

EXPERIENCE

Engineering Manager, Application Security April 2026 - Present

Twilio

Staff Product Security Engineer March 2024 - March 2026

Twilio

- Lead strategic security initiatives across multiple product teams and engineering organizations
- Architect and implement enterprise-scale security frameworks and automation solutions
- Drive security culture transformation through mentoring, training, and cross-functional collaboration

Senior Product Security Engineer September 2022 - March 2024

Twilio

- Designed and developed Snyk Autofactory, an automation framework written in TypeScript on the Serverless framework, AWS Lambda and ECS. It imports GitHub repositories from over 130 GitHub organizations into Snyk, collects Snyk security metrics, archives Snyk projects for archived GitHub repositories, and synchronizes Snyk organizations with GitHub organizations.
- Designed and developed the Product Security Metrics Framework (Multivac), enabling security engineers to build custom security metrics and deploy them on AWS (Lambda, ECS), with dashboards in Snowflake, Tableau or Domo.
- Mentor new and current teammates weekly on software development, product security processes, and design reviews.

Product Security Engineer July 2021 - September 2022

Twilio Segment

- Led the security review process for Segment's Data Purge feature, organizing and performing multiple threat models and managing security engineers.
- Upgraded Data Inventory (Retool application, Slack bot and security metrics cron job, built with Node.js) to be self-service, bridging the gap between data store ownership and engineering teams. It won the "Dino Stomp" award at Segment's 2021 R&D demo day and was demoed at the Twilio-wide engineering operations review.
- Helped design and build a security metrics framework to determine threat model coverage across Twilio.
- Helped design the Singular Security Review (SSR), a single point of entry for all security reviews that compiles engineering intake and automatically associates the required security teams through a custom JIRA workflow.
- Became Twilio's Snyk liaison: worked with the Snyk team, helped engineering adopt Snyk, and built custom Snyk tooling and security metrics.
- Triaged and validated Bugcrowd submissions and assigned them to the proper engineering team.
- Demoed around a dozen times at internal Information Security demo days.

Senior Application Security Engineer September 2019 - July 2021
Softrams

- Performed web application static code analysis and dynamic penetration testing.
- Developed application security assessment reporting standards and templates.
- Researched and developed internal security tools to optimize team performance.
- Delivered application security training (brown bags, lectures, hands-on labs); led technical interviews for developer and security candidates.
- Lead developer of Bulwark, an open source organizational asset and vulnerability management tool.

Senior Software Developer July 2018 - August 2019
Softrams

- Led a software development team responsible for functional execution of sprint goals.
- Built backend microservices with Node.js and the Loopback framework, and front ends with Angular 4 and TypeScript.
- Performed quarterly static and dynamic web application security assessments; developed assessment reporting standards and templates.
- Maintained a coding assignment for candidate interviews and led technical interviews; assisted with developer security training.

Application Security Consultant November 2015 - July 2018
nvisium

- Built front-end implementations of internal tools (TypeScript, AngularJS, Angular) and a cloud-based IDE using Monaco Editor.
- Performed secure code reviews, web penetration testing, and SDLC consulting.
- Researched offensive and defensive security techniques; contributed to open source security projects and training content.

Software Engineer December 2013 - November 2015
General Dynamics Mission Systems

- Developed and delivered user stories on a scrum team; built a single page application using Java, AngularJS, Bootstrap and HTML5.
- Supported software development reviews, scrum estimation, and CDRL document development with a systems engineering team.

SKILLS

Security: Application Security, Product Security, Threat Modeling, Static Application Security Testing (SAST), Secure SDLC, Security Metrics, Vulnerability Management, Risk Assessment, Compliance, Vendor Security, Security Reviews, Security Training, OWASP, SAMM, SLSA

Tools: Snyk, Semgrep, GitHub Advanced Security, Dependabot, TruffleHog, Burp Suite Pro, Bugcrowd, AssetNote

Engineering: TypeScript, JavaScript, Node.js, Python, Java, Angular, HTML5, CSS3, AWS (Lambda, ECS), Serverless, Docker, Buildkite, Git, GitHub, MySQL, PostgreSQL

OPEN SOURCE AND PROJECTS

- **Bulwark** (github.com/softrams/bulwark; TypeScript, 187 GitHub stars): organizational asset and vulnerability management tool with Jira integration, for generating application security reports. I was lead developer.
- **guppy-agent** (github.com/alejandrosaenz117/guppy-agent; TypeScript): AI-powered security reviewer that monitors pull requests for vulnerabilities and blocks merges.
- **bonfires-marketplace** (github.com/alejandrosaenz117/bonfires-marketplace): marketplace of Claude Code plugins for adversarial security and architectural code review.

- **fetch-cwe-list** (github.com/alejandrosaenz117/fetch-cwe-list; JavaScript): Node.js module to fetch, extract, and parse the latest or a specific CWE list from MITRE.
- **EXTREMIS** (playextremis.io; TypeScript, Three.js): playable top-down tactical defense game with procedural maps, enemy bots that scale to the player, synthesized WebAudio sound, mobile controls, and online play on Cloudflare Workers.

SPEAKING AND PUBLICATIONS

- DevSecCon 2024: From Novice to Catalyst: Scaling Security Tools with Metrics
- Twilio Blog (December 2024): Ten Years of Bug Bounty Programs: Updates to Twilio's Approach
- Absolute AppSec, Episode 247 (June 2024)
- Bulwark featured in the TL;DR Sec newsletter (#50, 2020)

EDUCATION

Virginia Commonwealth University - B.S., Computer Science August 2009 - December 2013